lowThrust — how it works

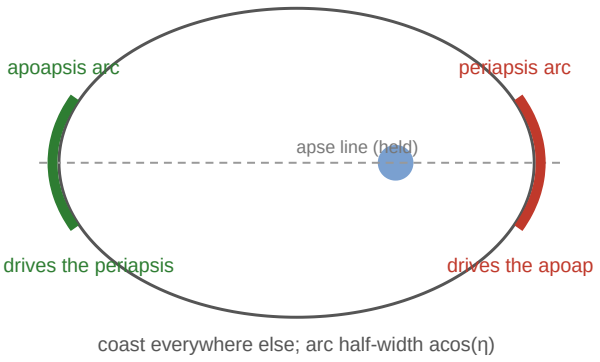
Low-thrust transfers within one SOI, to a moon, to another planet or a planet's moon: the solver, its phase machine and the stand-in solar system it flies in. No kRPC side yet.

What it does

Given the vessel's current orbit and a target orbit (apoapsis, periapsis, inclination, longitude of the ascending node, about any body; anything left out is free), it decides every tick whether to thrust, which way and how hard, until the target orbit is reached. Every decision is made from the current state and a fresh prediction; there is no stored schedule, so a weak or lost tick is made up by the same law later.

module	role
`orbit.py`	elements of any conic, exact propagation (universal variables), time to periapsis, Lambert, radius toward a fixed direction
`guidance.py`	the within-SOI law: `lock` (apse line and target plane), `command`, `arrived`
`plan.py`	flies the law in the simulator to pick the arc width that meets a deadline
`sim.py`	single-body stand-in; the vessel (constant mass, thrust, Isp); shared RK4 step and tick
`bodies.py`	stock bodies on Keplerian rails with their spheres of influence
`transfer.py`	targeting (closest approach, B-plane miss, correction, moon push, window, parking orientation, Lambert re-target, hazards) and the capture law
`mission.py`	the phase machine of a whole transfer in patched conics; routing and leg chaining
`tools/`	`simfly.py` (one SOI, deadline), `simmission.py` (a transfer: timeline, burns, dv), `sweep.sh` (destination × thrust × date grid)

The within-SOI law: Hohmann in many small burns



Two arcs centred on a fixed apse line; each fixes the opposite apsis.

Decoupled. A tangential burn at periapsis moves the apoapsis and, to first order, not the periapsis, and the other way round. The quantity driven is the radius toward the far end of the apse line, r = p / (1 + e·u), smooth through circular; its rate per m/s of prograde is differenced numerically (`RATE_DV`).

Arcs are ±acos(η) wide, η the off-apsis effectivity (default 0.99, ±8°). A burn starts half its remaining duration before the apsis, runs at full throttle until the radius is on target or the arc ends; the last tick's throttle is the fraction still needed. Prograde or retrograde only, so plain SAS modes can hold it.

Order: outward first (raise the apoapsis before the periapsis), inward first when lowering; the apoapsis is not lowered while the plane still has to change.

Apse line: held through each burn, re-locked to the osculating periapsis while coasting; a target with a held line (the parking ellipse) keeps it fixed and corrects it at quadrature once the apsides are close.

Plane change

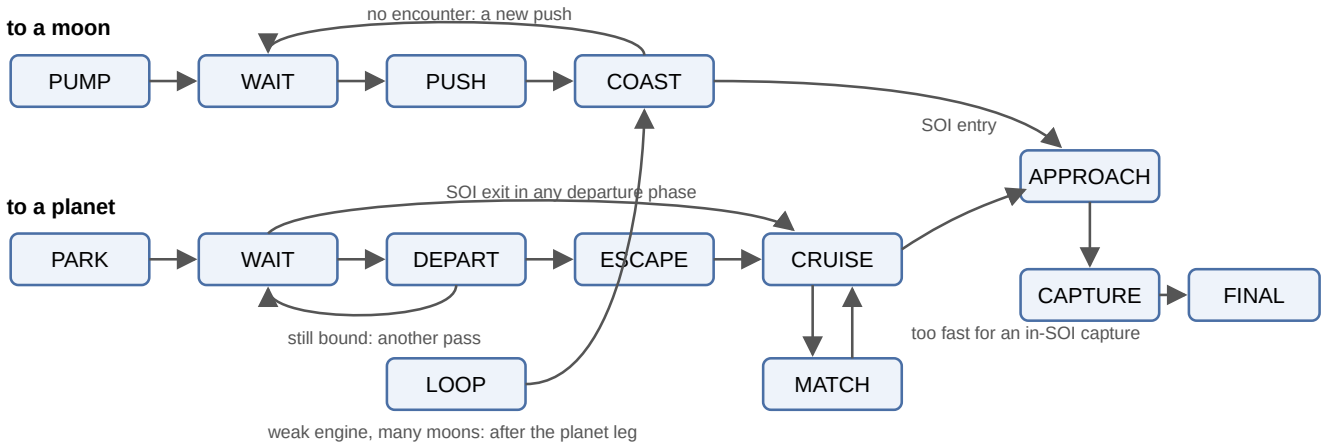
The target plane is fixed once (`lock`). Normal/anti-normal arcs of the same width at the two nodes with it, burn time $\delta \cdot h/(r_{node} \cdot a)$; a node whose radius is below η of the other's is skipped (a plane change costs the speed it turns, least where the vessel is slowest).

gate	effect
apoapsis still being raised	no plane burns yet: the plane is changed on the biggest orbit
plane still to change	the apoapsis is not lowered
periapsis still to change (measured as the apsis arc measures it — toward a held apse line when there is one)	plane burns only at the node on the apoapsis side (within an arc of quadrature allowed), where they share the periapsis burn
apsis arc and node arc both due	one burn: direction (prograde, normal) in proportion to the seconds each still needs, started half the combined duration early

The planner and the time budget

`plan.plan` flies the law in the simulator at `PLAN_GRID` (13) arc widths between `ETA_MAX` (0.999, ±2.6°) and `ETA_MIN` (0.2, ±78°) and takes the narrowest — cheapest — that arrives within the budget; without one it flies `ETA_DEFAULT`. The prediction is the law itself flown forward.

Transfers between bodies: the phase machine



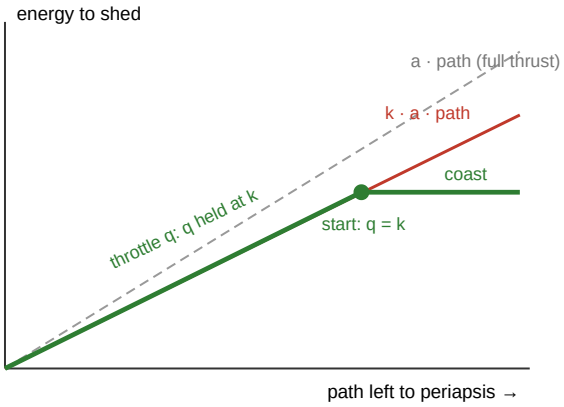
TRIM (a phasing burn at periapsis) and an evasive burn in an unplanned SOI can interrupt any phase. A planet's moon is two legs.

The phases of a transfer (`mission.Mission`), each re-decided every tick from the current state.

phase	what it does	ends when
PUMP	within-SOI law to the waiting orbit: in the moon's plane, apoapsis 0.6 of the way, periapsis as low as possible but high enough for the engine to capture in one pass and outside other moons' bands. With another moon inside the target's, the whole waiting orbit stays below that moon's band, periapsis raised until the push across it fits one pass.	reached
WAIT (moon)	once a revolution, ahead of periapsis: solve the prograde push there that puts a direct encounter (within 0.75 revolution) on the aim point. Skip the pass if its coast meets another moon, or if it arrives faster than the capture takes (the bar rises 25% per skip).	a push exists, periapsis half its duration away
PUSH	prograde on a ledger (planned dv less delivered), refined by local re-solves	ledger spent
COAST / CRUISE	B-plane corrections when one needs more than 0.5 m/s, rechecked every tenth of the time left. To a moon, no encounter goes back to WAIT. To a planet, a miss beyond min(20 SOI, 1% of the target's distance) or a correction over 150 m/s is re-planned as a Lambert arc; the encounter is looked for round the planned arrival. Triggers MATCH when planned.	SOI entry
PARK	within-SOI law, held apse line, to the parking ellipse oriented on the window's departure asymptote	reached
WAIT (planet)	coast to the periapsis pass nearest the window; skip a pass whose departure burn, flown forward on RK4 as DEPART will fly it, meets a moon	half the push before periapsis
DEPART	prograde until the hyperbolic excess is the window's, or 60° past periapsis (signed anomaly)	escaping: ESCAPE; still bound: WAIT
ESCAPE	coast out; a bound orbit whose periapsis fell below the floor is re-parked	SOI exit → CRUISE
MATCH	brakes about the star onto an aim point just outside the destination's SOI (below)	600 s before the aim point → CRUISE
APPROACH	nudge the hyperbola's periapsis onto the capture altitude along its gradient; coast an unplanned flyby	the capture law says burn
CAPTURE	the capture law (below); coasts through another body's SOI and re-decides on exit	bound (FINAL) / lost (fails)
LOOP	a weak engine's route to a moon of a many-moon planet (below): once a revolution, near apoapsis, the tangential burn that puts the next periapsis pass on the moon's aim point, centred on apoapsis; a phasing trim if the coming revolution meets another moon	burn done → COAST; no encounter → LOOP
FINAL	within-SOI law to the target orbit	arrived

Every orbit-shaping phase widens its arcs (cutoff down to 0.5, ±60°) until the apsis changes still to make fit `ARC_PASSES_MAX` (30) passes. **Floor guard:** outside CAPTURE and DEPART, a bound orbit whose periapsis is inside the atmosphere or surface is lifted along the periapsis gradient before anything else. **Thrusting ticks** change the velocity by at most `THRUST_DV_SHARE` (2%) of it.

The capture law



q = energy to shed / (a × path left to periapsis). Coast until q reaches k, then throttle at q: q stays constant and the last of the energy goes at periapsis.

The capture only has to make the orbit **bound**: below 0.3 of the SOI (or the final apoapsis, if higher), or, once past periapsis, anything below the highest apoapsis clear of the planet's moons — and always with the periapsis above the safe floor (30 km over the atmosphere). FINAL then shapes the orbit.

Short: if what is left fits within 80% of the time spent within ±60° of periapsis, the burn is centred on periapsis at full throttle.

Long: thrust along the velocity removes specific energy at exactly *a* per metre of path flown, wherever it is flown. So the law coasts until the energy above a bound orbit reaches `CAPTURE_PATH_SHARE` (k = 0.8) of what full thrust can remove on the path left to periapsis, then throttles at that ratio — a steady throttle near k, the rest in reserve. Outbound and still unbound: full throttle.

Direction: retrograde (sized to one tick's dv), plus a periapsis-gradient term that holds the periapsis between the floor and 1.5× the aimed one (`CAPTURE_RP_BAND`); below the floor, only the lift. Steps change the velocity by at most 5%.

It runs only inside the destination's SOI, where the destination alone pulls. Planning flies the same law (`transfer.fly_capture`) from the window's arrival to choose between a capture and MATCH.

MATCH: braking about the star

When: planned when the arrival is faster than a capture starting within 0.8 of the SOI can take. Its arrival speed v_s is 0.8 (`ARRIVE_MARGIN`) of the fastest the capture takes — a long capture inside the SOI, or one pass high and loose, whichever is faster.

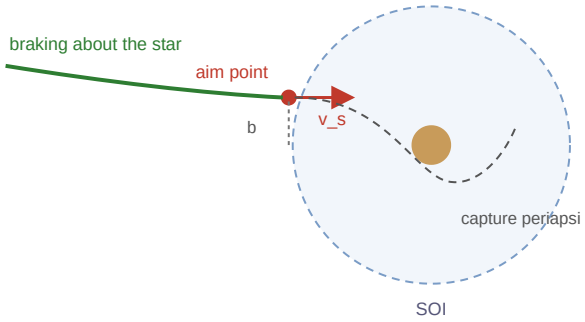
Start: from the cruise's predicted encounter (arrival speed v, time to closest approach): when braking from v to v_s at `MATCH_SHARE` (0.8) of the engine, plus the standoff, needs all the time left.

Law: zero-effort-miss / zero-effort-velocity guidance, $a = 6\,ZEM/t^2 - 2\,ZEV/t$. ZEM and ZEV come from coasting the vessel and the destination t ahead about the star. t is the time a constant deceleration from v takes to reach the aim point, $t = 2\,(v \cdot t_{ca} - d_{back}) / (v + v_s)$, recomputed every tick from a fresh encounter.

Aim point: `MATCH_STANDOFF` (1.05) SOI radii out, moving at v_s along the arrival direction, offset by the impact parameter of the capture periapsis on the B-plane side the cruise aims for.

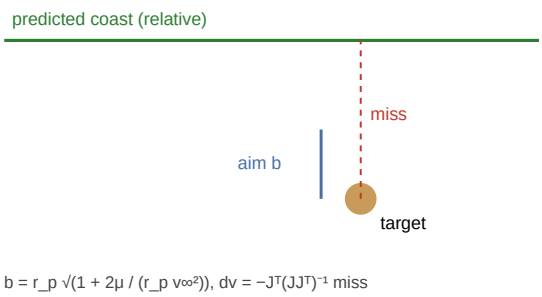
Moons: a moon leg plans MATCH too, in the planet's frame, with v_s from the moon's capture limit — at least a quarter of the escape speed at the moon's SOI edge (slower is falling in from rest). A new push clears the remembered encounter.

End: 600 s before the aim point (`MATCH_END_S`), or once $v \le 1.05\,v_s$; CRUISE then carries it into the SOI.



MATCH ends at the aim point at v_s with the capture's impact parameter b; the capture law takes over inside.

Targeting



$b = r_p \sqrt{(1 + 2\mu / (r_p v_\infty^2))}$, $dv = -J^T(JJ^T)^{-1}$ miss

The B-plane miss and the minimum-norm correction.

Closest approach of the exact two-body coast to the target on its rails: a coarse scan then golden section, or a window round a given UT (the planned arrival). A minimum at the window's edge is no encounter.

Aim point: the impact parameter b for the capture periapsis, on the side whose passage is prograde about the target's orbit normal. **Correction:** numerical Jacobian of the two miss components against the three impulse axes, minimum-norm.

Moon push: the smallest prograde impulse at periapsis whose signed closest-approach distance equals b, searched up to `PUSH_DV_FACTOR` (2) times the tangential push that reaches the moon's orbit. A pass arriving faster than the capture takes is skipped (the bar rises 25% a skip). **LOOP's burn:** the same root search on the apoapsis burn, over the range that shifts the next periapsis by about one moon period. **Re-target:** Lambert to the aim point over arrival times 0.6–1.6 of the planned time left, cheapest in impulse plus arrival speed, held while burning.

Departure, capture design and hazards

Window: Lambert over departure dates (one synodic period from the pump's estimated end) and flight times (0.5–1.5 of the Hohmann's), least $|v_\infty$ out$|$ + $|v_\infty$ in$|$. **Parking ellipse:** periapsis above the atmosphere, apoapsis 0.9 of the highest clear of the planet's moons; its plane contains the departure asymptote, the one cheapest to reach. A weak engine's departure takes several periapsis passes; whatever v_∞ it leaves with, the cruise's re-target builds the rest about the star.

A planet with moons: a tight capture (apoapsis under the innermost moon) is compared, by predicted dv, with a loose one (0.5 SOI, low periapsis) plus the impulsive cost of lowering it at periapsis afterwards; the cheaper is planned.

A moon of a many-moon planet, weak engine (LOOP): when the planet leg would have to capture loosely for want of a feasible tight capture, it captures with its periapsis on the target moon's orbit and matches only the moon's plane; the moon leg then starts in LOOP instead of PUMP.

Capture design: the lowest capture periapsis (≥ the target's, ≥ the floor) at which one pass sheds the arrival speed into the preferred ellipse (0.3 SOI, under the moons) or a looser one; failing that, a long capture inside the SOI if the planning prediction says it starts within 0.8 SOI; failing that, MATCH. Refused only if even an arrival at rest cannot be captured.

Hazards: each arc burn's pass dv is capped at the largest fraction whose next revolution stays out of every other moon's SOI; a phasing trim of ±0.5–8 m/s at periapsis if the coming revolution would meet a moon; an evasive burn along the periapsis gradient inside an unplanned SOI with the periapsis below the floor; after a slingshot, braking until the apoapsis is inside 0.6 SOI.

The stand-in

Patched conics: the vessel's state is relative to the body whose SOI it is in. A coasting tick is propagated exactly (universal variables, starting from whichever of the short- and long-arc guesses is closer) and bisected to an SOI boundary when it crosses one; the frame then switches by the body's state on its rails. Thrusting ticks are `THRUST_SUBSTEPS` (4) RK4 steps. Ticks are uniform in angle (r^2/h per 2°, capped at period/180). The vessel has constant mass and unlimited propellant — the stand-in plans paths; dv is thrust/mass × burn time and xenon is reported from it with the rocket equation. A flight ends on arrival, impact, the time limit (`MISSION_MAX_DAYS`) or a failed capture.